

EMCH 367 Controls

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Fall 2024

Exam #3

Problem 1. (20pt) True or False

1. $\tan^{-1}(1/1) = \tan^{-1}(-1/-1)$
2. If $\angle G(j\omega) > 0$, we say that the response is leading the input.
3. Consider the compensator $G_c(s) = K(s+a)/(s+b)$. If $b > a$, then it is a lead compensator.
4. For Bode plot, the magnitude is computed using $10 \log |G(j\omega)|$.
5. The phase contribution of $G(s) = j\omega$ is $+90^\circ$.
6. For $G(s) = 1/(Ts+1)$, the difference between the asymptote and the exact curve at the corner frequency is 10 dB.
7. High pass filter blocks the signal with high frequency.
8. The bandwidth frequency is the lowest frequency for which the gain has declined 3 dB below the asymptote level $G(0)$.
9. For Nyquist stability criterion, Z means the number of zeros of $1+G(s)H(s)$ in the LHP.
10. For Nyquist stability criterion, N means the number of clockwise encirclements of the $-1 + j0$ point

Number	T/F	Number	T/F
1	F	6	F
2	T	7	F
3	T	8	T
4	F	9	F
5	T	10	T

Problem 2. (20pt) Consider

$$G(s) = \frac{(s+1)(s+5)}{s(s^2+4s+10)(s^2+8s+25)}.$$

Find $y_{ss}(t)$ for $u(t) = \sin(2t - 1)$.

Solution: First, note that

$$G(j\omega) = \frac{(1+j\omega)(5+j\omega)}{j\omega(10-\omega^2+j4\omega)(25-\omega^2+j8\omega)}.$$

For $u(t) = \sin(2t - 1)$, we have

$$G(j2) = \frac{(1+j2)(5+j2)}{j2(6+j8)(21+j16)},$$

and hence, it follows that

$$\begin{aligned}\angle G(j2) &= \angle(1+j2) + \angle(5+j2) - \angle(j2) - \angle(6+j8) - \angle(21+j16) \\ &= \tan^{-1}\left(\frac{2}{1}\right) + \tan^{-1}\left(\frac{2}{5}\right) - \tan^{-1}\left(\frac{2}{0}\right) - \tan^{-1}\left(\frac{8}{6}\right) - \tan^{-1}\left(\frac{16}{21}\right) \\ &= -1.6615.\end{aligned}$$

Next, the magnitude is given by

$$\begin{aligned}|G(j2)| &= \frac{\sqrt{5}\sqrt{29}}{2\sqrt{100}\sqrt{441+256}} \\ &= 0.0228.\end{aligned}$$

Therefore, the steady state response $y_{ss}(t)$ is given by

$$y_{ss}(t) = 0.0228 \sin(2t - 2.6615).$$

Problem 3. (40pt) Consider the following uncompensated system.

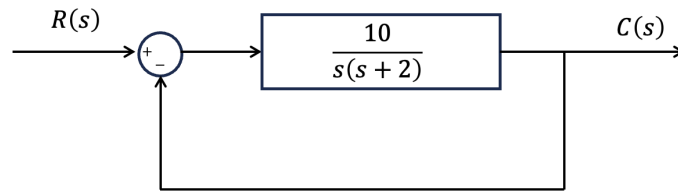


Figure 1

(a) (5pt) Identify the closed loop poles.

Solution: Note that the closed loop transfer function is given by

$$\begin{aligned}\frac{C(s)}{R(s)} &= \frac{\frac{10}{s(s+2)}}{1 + \frac{10}{s(s+2)}} \\ &= \frac{10}{s^2 + 2s + 10},\end{aligned}$$

and hence, the closed loop poles are located at

$$s = -1 \pm 3j.$$

(b) (5pt) Let the target natural frequency and damping ratio are $\omega_n = 3$ rad/s and $\zeta = 0.5$, respectively. Identify the target dominant closed loop poles.

Solution: To have $\omega_n = 3$ rad/s and $\zeta = 0.5$, the characteristic equation should include

$$s^2 + 3s + 9,$$

and hence, the dominant poles should be at

$$s = -1.5 \pm 2.5981j.$$

(c) (20pt) To achieve the goal, a lead compensator

$$G_c(s) = K \frac{s + a}{s + b}, \quad b > a,$$

is used as follows. Find a and b .

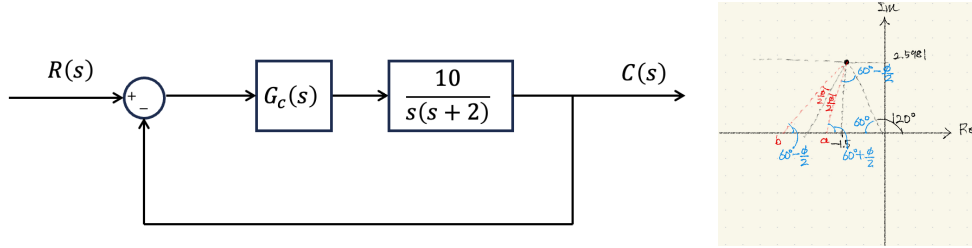


Figure 2

Solution: The phase of the uncompensated open loop at the target dominant poles is given by

$$\begin{aligned} \angle G(s)|_{s=-1.5+2.5981j} &= -\angle(-1.5 + j2.5981) - \angle(0.5 + j2.5981) \\ &= -199.1067^\circ \\ &= 3.4751 \text{ rad.} \end{aligned}$$

Therefore, the contribution of the lead compensator should be $\phi = 19.1067^\circ = 0.3335 \text{ rad}$. It follows from sine law that

$$\begin{aligned} a &= 3 \frac{\sin(60^\circ - \phi/2)}{\sin(60^\circ + \phi/2)} = 2.4686, \\ b &= 3 \frac{\sin(60^\circ + \phi/2)}{\sin(60^\circ - \phi/2)} = 3.6456. \end{aligned}$$

(d) (10pt) Find K .

Solution: Now, note that

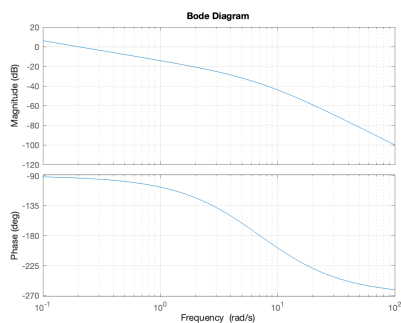
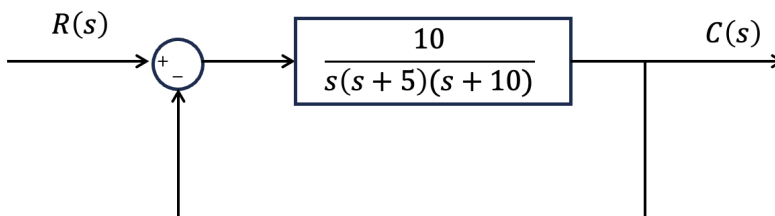
$$\left| K \frac{10(s+a)}{s(s+2)(s+b)} \right|_{s=-1.5+2.5981j} = 1,$$

and hence,

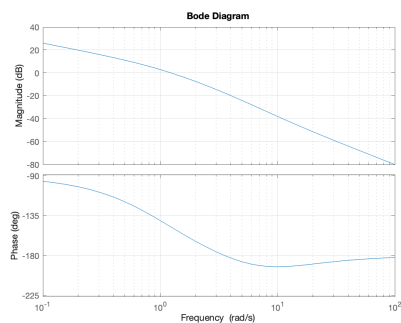
$$K = \left| \frac{(s+b)s(s+2)}{10(s+2.4686)} \right|_{s=-1.5+2.5981j} = 0.9646.$$

Problem 4. (20pt)

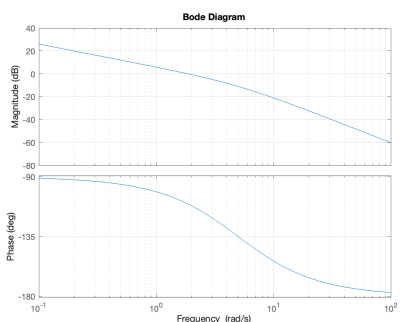
(a) (10pt) Choose the correct Bode plot. (Justify your answer)



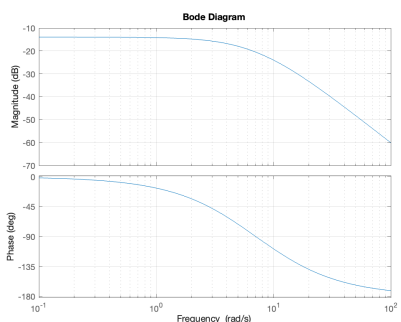
(a)



(b)



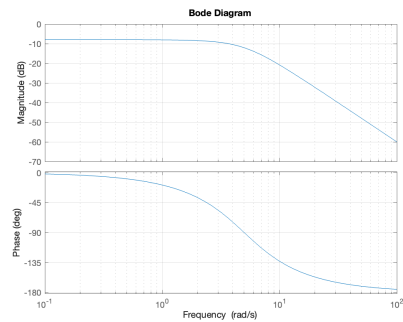
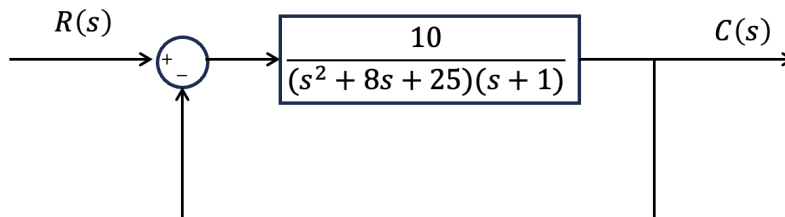
(c)



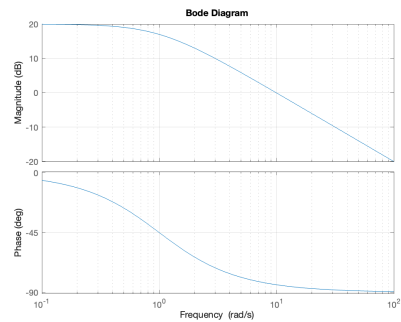
(d)

Solution: (a). The magnitude has to be -60 dB/decade after 10 rad/s. The phase should be -270° at the end.

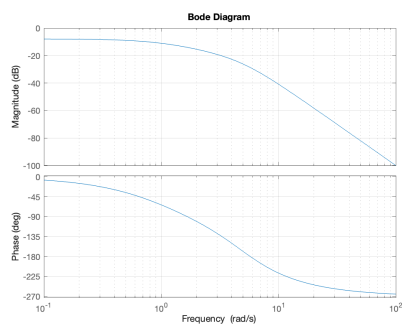
(b) (10pt) Choose the correct Bode plot. (Justify your answer)



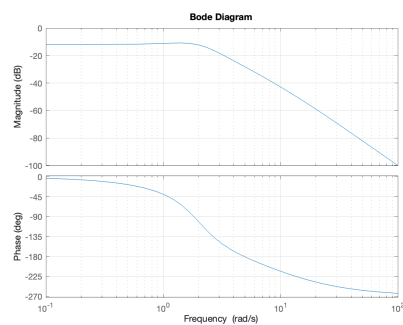
(e)



(f)



(g)



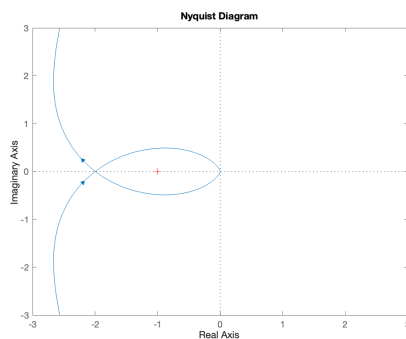
(h)

Solution: (g). The magnitude should be -60 dB/decade after 5 rad/s. The phase should be -270° at the end. (h) has an increase in the magnitude around 2 rad/s.

Bonus Problem (20pt) Consider the following open loop transfer function

$$G(s) = \frac{2}{s(s^2 + s + 1)}.$$

The Nyquist plot is given by



Please explain why the Nyquist plot is given as the above.

Hint: Use the following Nyquist path.

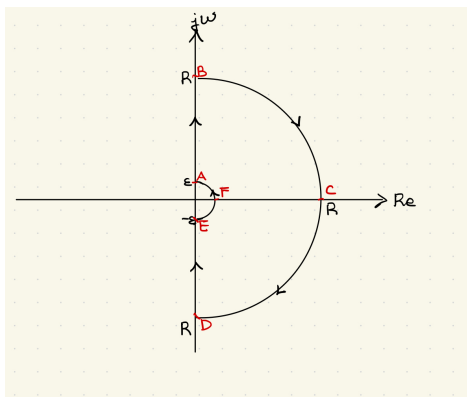


Figure 3

Then, identify $|G(j\omega)|$ and $\angle G(j\omega)$. Do the analysis for $A \rightarrow B$, $B \rightarrow C \rightarrow D$, $D \rightarrow E$, and $E \rightarrow F \rightarrow A$.

Solution: First, note that

$$G(j\omega) = \frac{2}{j\omega(1 - \omega^2 + j\omega)},$$

and hence, we have

$$|G(j\omega)| = \frac{2}{|\omega|\sqrt{(1 - \omega^2)^2 + \omega^2}},$$

$$\angle G(j\omega) = -\tan^{-1}(\omega, 0) - \tan^{-1}(\omega, 1 - \omega^2).$$

For the first segment, $(0, j\varepsilon) \rightarrow (0, jR)$, note that

$$|G(j\varepsilon)| = \frac{2}{\varepsilon\sqrt{(1 - \varepsilon^2)^2 + \varepsilon^2}} \rightarrow \infty,$$

$$\angle G(j\varepsilon) = -\tan^{-1}(\varepsilon, 0) - \tan^{-1}(\varepsilon, 1 - \varepsilon^2) \rightarrow -90^\circ - 0 = -90^\circ,$$

$$|G(jR)| = \frac{2}{R\sqrt{(1 - R^2)^2 + R^2}} \rightarrow 0,$$

$$\angle G(jR) = -\tan^{-1}(R, 0) - \tan^{-1}(R, 1 - R^2) \rightarrow -90^\circ - 180^\circ = -270^\circ.$$

For the next segment, $(0, jR) \rightarrow (R, 0) \rightarrow (0, -jR)$, note that for $s = Re^{j\phi}$, $\phi : 90^\circ \rightarrow 0^\circ \rightarrow -90^\circ$,

$$G(Re^{j\phi}) = \frac{2}{Re^{j\phi}(R^2e^{j2\phi} + Re^{j\phi} + 1)},$$

and hence,

$$|G(Re^{j\phi})| \rightarrow 0, \quad \angle G(Re^{j\phi}) : -270^\circ \rightarrow 0 \rightarrow 270^\circ.$$

Next, for $(0, -jR) \rightarrow (0, -j\varepsilon)$, note that

$$|G(-jR)| = \frac{2}{R\sqrt{(1 - R^2)^2 + R^2}} \rightarrow 0,$$

$$\angle G(-jR) = -\tan^{-1}(-R, 0) - \tan^{-1}(-R, 1 - R^2) \rightarrow -(-90^\circ) - 180^\circ = -90^\circ,$$

$$|G(-j\varepsilon)| = \frac{2}{\varepsilon\sqrt{(1 - \varepsilon^2)^2 + \varepsilon^2}} \rightarrow \infty,$$

$$\angle G(-j\varepsilon) = -\tan^{-1}(-\varepsilon, 0) - \tan^{-1}(-\varepsilon, 1 - \varepsilon^2) \rightarrow -(-90^\circ) - 0 = 90^\circ.$$

Finally, for small semicircle, $s = \varepsilon e^{j\phi}$, $\phi : -90^\circ \rightarrow 0^\circ \rightarrow 90^\circ$, note that

$$G(\varepsilon e^{j\phi}) = \frac{2}{\varepsilon e^{j\phi}(\varepsilon^2 e^{j2\phi} + \varepsilon e^{j\phi} + 1)},$$

and hence,

$$|G(\varepsilon e^{j\phi})| \rightarrow \infty, \quad \angle G(\varepsilon e^{j\phi}) : 90^\circ \rightarrow 0 \rightarrow -90^\circ.$$

Therefore, we have the Nyquist plot that looks like Figure 4.

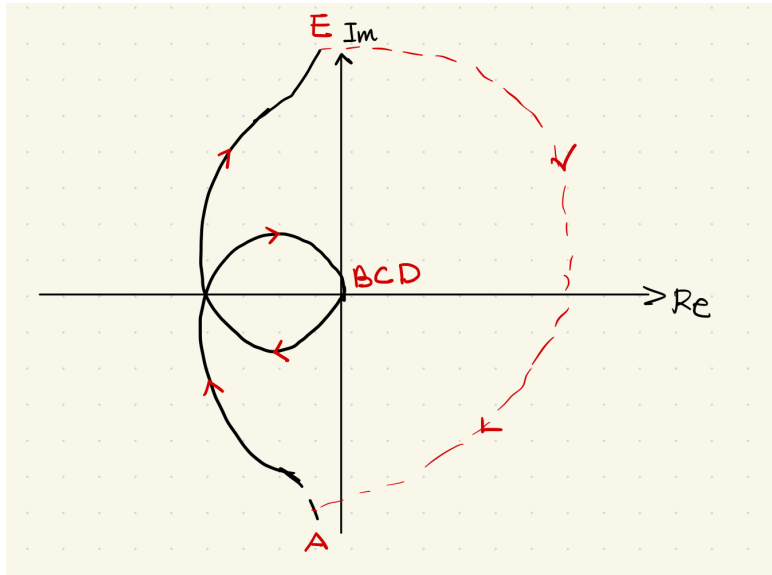


Figure 4